

Lab 3

PUBH 523/623

Your name here!

Warning

Note that this is the editing file. **There are detailed instructions on our course page for Lab 3 that are not in this file.**

Lab activities

Restate your selected project

Task

Copy and paste the row from the above table that corresponds to your project into your `.qmd` file. This will help you (and me hehe) keep track of your project question, population, main variables, and composite score(s) as you work through this lab.

Import your saved data from Lab 2

Task

From the `.rda` file you exported at the end of Lab 2, import your data into R using `import()` from the `rio` package. Save this as a new object called `wbns_05_composite`. Print out the number of rows and columns to verify the import.

Quick summaries of your variables

! Task

Take your `wbns_05_composite` dataset and use `skim()` to generate a quick summary of your main variables and composite score(s). Save the result as a new object called `quick_summaries`, and print it out. Use this to check for any unexpected values, missing data, or other issues that may need to be addressed before moving forward.

Creating categorical versions of your composite score(s)

! Task

Use `mutate()` with `case_when()` or `cut()` to create a new categorical variable for each composite score, with meaningful category labels. Save the result as a new object called `wbns_06_cat`. Use `tabyl()` or `count()` to display how many respondents fall into each category, and confirm no one was left out (e.g., turned into an unexpected NA).

Building a summary table

! Task

Use `tbl_summary()` to build a clean overall summary table of your composite score(s) and other main variable. Save this as a new object called `summary_table_overall`, and display it.

Customize at least one thing about the table's appearance or labels (e.g., with `modify_header()` or by relabeling variables) so it's easier for someone unfamiliar with your project to read.

Grouped summaries

! Task

Use `group_by()` and `get_summary_stats()` to calculate your composite score's common summary statistics separately for each level of your chosen grouping variable. Write 2-3 sentences interpreting what you see. Does the composite score noticeably differ across groups? Does this align with what you'd expect given your project question?

Note

Optional/Additional: You can also pass your grouping variable into `tbl_summary(by = ...)` to get a nicely formatted side-by-side table instead of (or in addition to) the `group_by()/get_summary_stats()` output above.

Export tables

Task

Save `wbns_06_cat` and `summary_table_overall` in a `.rda` file in your `data_project` folder. Name the file something that will help you identify it later, like `wbns_data_lab_03_work.rda` or `wbns_data_03_summ.rda`.